

Accor Assessment

Question 1

Compute & Architecture Design a containerized infrastructure on AWS EKS. Your solution must demonstrate how the application remains available and responsive during extreme load events and infrastructure failures.

Answer : In order to design a full scale high availability system we need to understand the comp flown every week, every month, during sale and promotions to handle and process the user request. Considering infrastructure load distribution, network, security, api server, kubernetes

User request

https request (TLS)

■ Route53 (DNS resolution)

■ Cloud Front

Application Web Firewall

API gateway

Application load balancer

Ingress

Service

EKS

API etcd Controller Scheduler
Server Manager

Pods on multi nodes and distributed on Multi Availability Zones for high availability.
Pod 1 | Pod 4 | AZ2 Pod 6 AZ3 - Node 3
Pod 2 | -Node 1AZ1 Pod 5 | Node2
Pod 3 |

Redis cache (for faster transactions to store static data)

RDS on AZ1 (primary)
-Standby RDS on AZ2

Worker node EC2 Instances
Kubelet
Nodes
Pods
Container Runtime
Containerd
HPA
Karpenter/Cluster Autoscaler

PDB (pod disruption budget) to be defined for high availability which defines the minavailable:2

Cloud Front : Stores cache data for faster performance, Absorbs excess traffic, Protects from D

AWF - Allows only those requests which are set to be allowed through firewall to ALB 443
It absorbs excess traffic protects SQL ingestions like multiple null pointers, Ddos attacks. M

ALB - Distributes the traffic requests

Request hits the API gateway on 8080 port where the gateway authenticates the request
on the exposed api

Ingress route the request from api gateway to the service based on the endpoint like
accor.com/search
accor.com/availability
accor.com/loyalty-points
accor.com/payment

Service distributes service request to the pods like a load balancer

HPA - Keeps a check on the CPU Target, lets say if we define threshold at 75%, HPA based on /m

Controller manager : Creates new replicas set and it creates new pods based on desired state vs

Liveness Probe - If the application is crashing and kubernetes restarts the container
Readiness probe - Checks if DB,nodes, pods are ready to take the traffic, if its failed, pods a

Schedule assigns the pods to the nodes based on the load and availability and based on Pod Affin
defined atleast 1 pod to be running on each node.

Cluster autoscaler or karpenter spins up additional nodes and places new pods whereas karpenter

Kubelet : Its a node agent than run the pod by calling container runtime where containerd will

Multi availability zone which is highly required for higher availability.
and due to cross region read replicas the availability remains high. on Pods, Nodes, RDS, S3 req

Define RTO (recovery time objective) - 10minutes
and RPO (Recovery Point Objective) (5mins) to sync database read replicas before any failure

Monitoring and Observability

Cloud Watch

Prometheus

Grafana

Datadog

Kafka is used for queue which uses broker which communicates and processes between producer and

During extreme traffic spikes, Redis reduces database load by querying static data and not hitting
HPA automatically scales application pods by creating replicaSet

Cluster Autoscaler or Karpenter provisions additional worker nodes if the node dies or CPU,Mem

During infrastructure failures, EKS reschedules pods to healthy nodes,

Multi-AZ deployment ensures application availability, and RDS Multi-AZ provides automatic data

Question 2

Question 2 - Scalability Strategy The system must automatically handle the 10x traffic spike so no manual intervention. Define your strategy for scaling both the application and the underlying infrastructure

Answer - Assuming normal rate of request is 100 requests/second and it spikes to 1000 request per second

Use predictive modelling based on the previous data by adding 20% additional count

We will have to assign Horizontal Pod Autoscaler which senses the CPU utilisation based on the metrics where the controller manager detects the change in desired state vs actual state and create replicas

If 6 pods are running to process 100 req/sec, HPA will scale it to 20 pods

By increasing the number of pods the node resource utilisation will also spike and pods might fail. Cluster autoscaler or karpenter will increase the nodes and add it to the cluster where the scheduler

Define CPU & Memory request vs limits

Redis cache - It stores the static information like loyalty points, search filters, redemption points to avoid latency and if not available it goes to the database

Multi Availability zones where the Nodes, Pods, RDS, S3 will have cross region read replicas.

Read replica can be used to distribute the DB request among primary/master and slave (read replicas)

CDN (Content driven network)/Cloud Front - Ddos attacks, caching api, images, redemption points

AWF - filters bot requests on specific api, protects SQL injections

Kafka - Queues are loaded where broker processes the queue between producer to consumer and fast

Question 3

Question 3 Security & Networking Design a network and security architecture that protects customer data and infrastructure integrity. Apply "Least Privilege" and "Defense in Depth" principles to the architecture.

Answer - Create a Virtual private cloud in AWS and define public and private subnets using terraform. Public subnets that are exposed to the internet such as AWF and ALB

Private subnets for EKS cluster, Redis, RDS as the internal data and user data should not be exposed to the internet

https TLS connections
Encrypted TCP packet

Route53 which resolves DNS

Cloudfront, Application web firewall

API gateway for authentication of requests and services

IAM for authentication and authorisation where Users, Roles, Groups and Policies

User request

https request (TLS)

■ Route53

■ Cloud Front /CDN

Application Web Firewall

API gateway

Application load balancer

Ingress

Service

EKS

Pods

■ Redis

■ RDS encrypted to access the database

IRSA will define the access for pods

Pod A - Read loyalty_points

IAM for authentication and authorisation where Users, Roles, Groups and Policies

Groups - Lets say developer and QA can have only READ access in PROD,

operations support and SRE can have only Admin access

Network rules allow traffic only from 443 and 8080

Adding multiple security layers where if cloud front and AWF is bypassed, we got ALB, IAM, IRSA

Least privilege is when IAM IRSA RBAC Security groups are defined where to limit pods access

Question 4

Question 4 - Reliability & Observability Define how you will measure the health of the service system will recover from failures (e.g., AZ outage, bad deployment) with minimal impact to the customer.

Answer - Use predictive modelling based on user request traffic

Define Error rate,SLA,SLO and SLI

SLO is the target error rate such as 99.5%

Maintain the SLI at 99.5 and above such as 99.8%

SLA is the agreement with customer where we are charged with penalties if crossed the threshold

Configure Cloudwatch using Alert Manager

Prometheus that collects the data from exposed /metric

Grafana where data is displayed in graphical format

Datadog

We configure this to capture CPU, Memory, latency, DB, pods, Nodes, Availability zones, logs, etc

We can use helm to install this on the agents

```
helm install datadog-agent/datadog/datadoghq
```

Define target utilisation of CPU

As soon as the resource utilisation crossed the threshold, Datadog/Grafana/Cloudwatch alerts trigger

Based on the alert we check the CPU and memory utilisation along with Network and disk latency,

Work with the engineering teams and Operations to understand errors and reduce the noisy alerts to keep it more focused.

Liveness Probe - If the application is crashing and kubernetes restarts the container

Readiness probe - Checks if DB,nodes, pods are ready to take the traffic, if its failed, pods and nodes both should be healthy and running

Reduce toil by automating the KPI service success/failure rate as a dashboard report instead of manual checks to make the system better and lesser error rate..

If AZ1 fails, failover happens with the help of Route53 & pods run on AZ2 and AZ3

If Node A running beyond CPU threshold HPA will add pods to healthy nodes to reduce CPU utilisation

If all the nodes crossed threshold, cluster autoscaler add new nodes to the cluster and assign pods

Add rollback procedures and rollout strategies.

```
kubectl rollout status deployment.yaml
```

```
kubectl rollout undo deployment.yaml
```

Use of Canary or Blue- Green deployment methods to check system performance and utilisation and make it primary
In Blue- Green we push the deployment in stand by server and make it primary

Question 5

Operations

- Day 2 Operations: Propose strategies to minimize operational toil.
- Task Assignment: Outline how the implementation tasks for this project should be assigned across a team of 3 engineers (1 Senior, 2 Juniors).

Answer : In order to reduce the operational toil we need to understand the scope of reducing the

We need to break it into 4 sections

IAC - Terraform

Deployments - Kubernetes and ArgoCD

Monitoring and observability - Datadog/Cloud Watch/Grafana

Find the opportunities of automating the manual efforts

Work with product, development engineering and IT ops teams to understand the manual efforts and

Infrastructure changes such as adding VPC,DB servers, Nodes to cluster, Patching updates, Adding new services can be automated using terraform

Deployments : Pod scaling, Node scaling, Restart containers, Restart the pod, failovers to Availability Zones can be reduced using HPA (Horizontal Pod Autoscaler), Karpenter or cluster autoscaler, failover can be automated using ArgoCD. Also once the code is pushed to github with a PR raised to running CI/CD pipelines running build and deployment. kubect apply -f deployment/payment can be automated using ArgoCD

Monitoring and observability: Understanding the end to end architecture and infrastructure of the application to capture CPU, Memory, Disk, Latency, Nodes health, Pod health of the infra resources.

Use endpoints flow to create traces, request rates, error rates,error logs, APM

Automate alerting based on the priority of the incident to slack,chat space,email, xMatters, On-call rotation

Setup dashboards based on business and operations KPI to enhance the system better based on errors and performance

Automate health check reports of infra resources bi-weekly to engineering and management.

Automate synthetic monitors and flap test that logs in with defined credentials, tests the application functionality fails.

Reduce noisy alerts

Integrating Bits Ai or Resolve Ai we can introduce to triage alerts, identify root causes, and suggest actions

Explain the entire flow and gaps to the team, take their ideas into considerations and based on the team's input

Senior Engineer will be understanding the architecture, checking the existing terraform into a single file, mentoring engineering teams, mentoring juniors.

Juniors Engineer 1 will be writing/updating deployment.yaml scripts to kubernetes and deployment.yaml scripts

Junior Engineer 2 will be assigned with Monitoring and Observability .

As a lead i will be working with Product, development, support, stake holders to understand the requirements